SOFTWARE
REQUIREMENTS
SPECIFICATIONS

18/08/2026

Sebenza Clocking System

Prepared for Umzamo Analytical Services

PROPOSAL BY

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PROJECT INTRODUCTION

Sebenza Clocking System is a proposed employee attendance and working-hours management system for Umzamo Analytical Services, a coal laboratory.

The project was identified from an existing attendance-management process in which employees manually record their clock-in and clock-out times and working hours using paper-based forms.

The current process relies on employees entering their own attendance information. According to the initial information provided by the stakeholder, there are situations where employees arrive late but record their normal expected starting time. There are also situations where overtime is recorded even though the employee did not work overtime.

This creates a need for a more reliable method of recording employee attendance and working hours.

The proposed Sebenza Clocking System will investigate and provide a digital method for recording employee clock-in and clock-out events using automatically generated timestamps. The system will also investigate an appropriate process for managing and verifying overtime.

The proposed solution will consist of a Flutter mobile application, an ASP.NET web application and a shared Supabase backend.

The project will be developed according to stakeholder requirements gathered from Umzamo Analytical Services.

PHASE 1: PLANNING, REQUIREMENTS AND FEASIBILITY

1. **STAKEHOLDER AND PROBLEM ANALYSIS**

1.1 ORGANISATION AND STAKEHOLDERS

Organisation: Umzamo Analytical Services.

Industry: Coal Mining Laboratory.

Primary Project Problem: Manual employee attendance and working-hour recording.

Proposed System: Sebenza Clocking System

Umzamo Analytical Services is the external stakeholder organisation for this project.

The organisation has been selected because its current employee attendance process is manually managed and relies on employees recording their own attendance and working-hour information on paper.

Stakeholder	**Role**
Employees	Record their attendance and working hours
Supervisors/Management	Monitor and verify employees' attendance and working hours
Human Resources	Manage attendance records
Finance	Use approved working-hours and overtime information
Project Team	Investigate, design, develop and test the proposed system

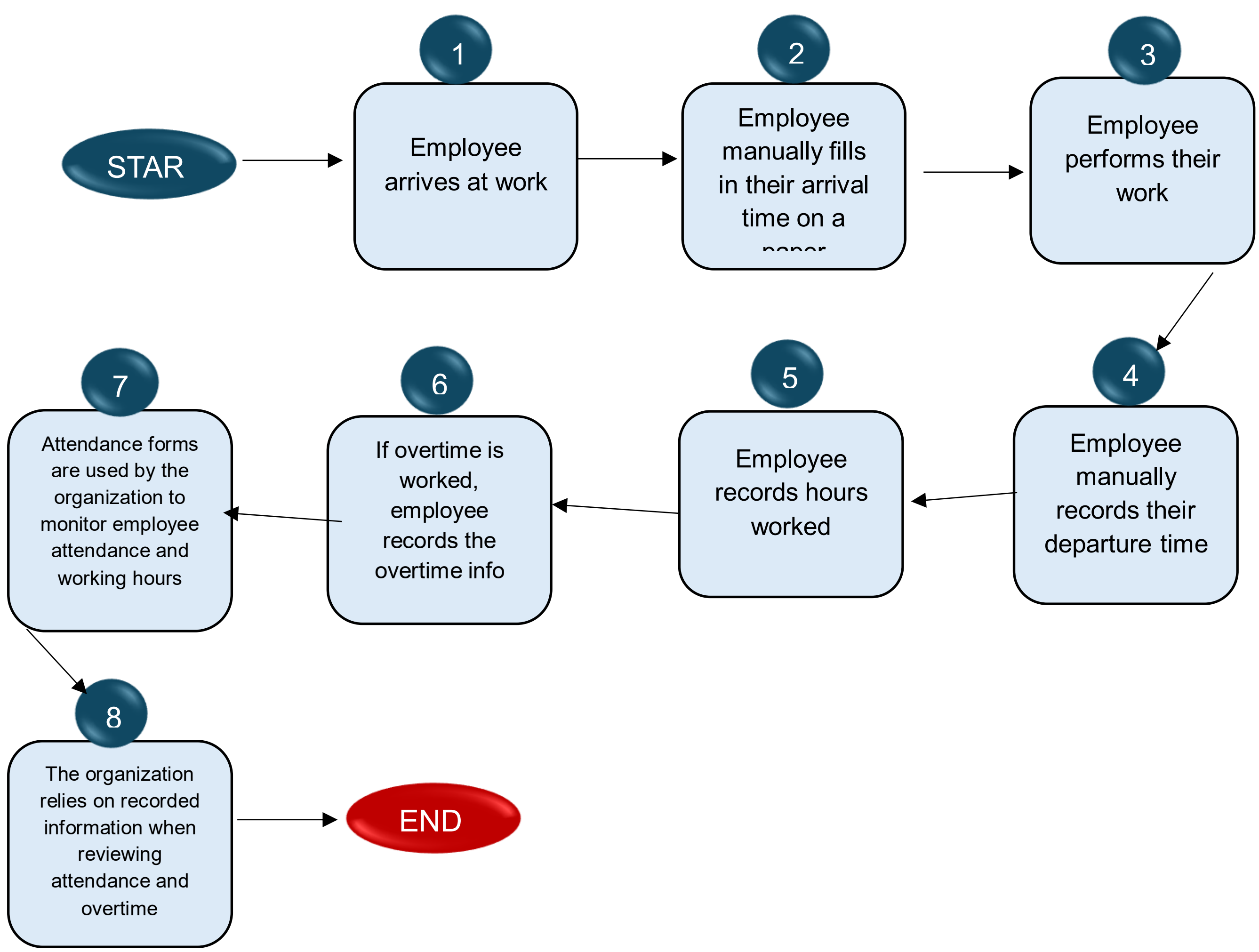
1.2 Current Process and People Involved

The current clocking in & out process is paper/manual based

The initial process is identified as follows:

1. Employee arrives at work
2. Employee manually fills in their arrival time on a paper attendance form
3. Employee performs their normal work
4. Employee manually records their departure time
5. Employee records hours worked
6. If overtime is worked, the employee records the overtime information
7. Attendance forms are used by the organisation to monitor employee attendance and working hours
8. The organisation relies on recorded information when reviewing attendance and overtime

1.2.1 Current Process diagram



1.3 Problem Statement and Affected Users:

Umzamo Analytical Services currently uses a manual paper-based process for recording employee attendance and working hours. Employees manually record their clock-in and clock-out times and may also record overtime.

The current process relies on self-reported information and does not automatically verify the actual time at which an employee arrives at or leaves the workplace. The stakeholder has identified instances where employees arrive late but record their normal starting time and instances where overtime is recorded even though overtime was not worked.

As a result, the organisation faces a risk of inaccurate attendance and working-hour records and may have trouble verifying the accuracy of recorded overtime.

A digital attendance management system is therefore required to investigate whether employee attendance can be recorded using automatically generated timestamps and whether overtime can be managed through a more controlled and verifiable process.

Effected Users;

- **Employees:** Employees are directly affected because they currently use paper forms to record their attendance and working hours.
- **Management:** Management may require attendance information to monitor employee attendance and working hours.
- **Payroll/Finance:** Payroll or finance personnel may use approved working-hour or overtime information if confirmed by the stakeholder.
- **Administration:** Administrative personnel may be responsible for managing employee attendance information.

1.4 Problem Experienced and its Consequence

The primary problem identified is that employees' attendance, working hours and overtime hours are manually recorded by employees

The Stakeholder has indicated that:

- Employees arrive late but record their normal starting time
- Employees take longer lunch breaks
- Employees may record overtime that was not actually worked
- The paper-based process does not automatically capture the actual hours worked and actual time at which an employee clocks in or out
- Attendance and overtime information therefore requires manual recording and verification

These consequences were confirmed by the organisation's stakeholders during the requirements investigation

Consequences of the Current problem:

- Inaccurate attendance records
- Difficulty verifying actual arrival time
- Difficulty verifying actual departure time
- Inaccurate working-hour information
- Incorrect overtime information
- Additional administrative work
- Difficulty maintaining reliable historical attendance records

1.5 Project Aim and Objectives

AIM:
The aim of the Sebenza Clocking System project is to develop an prototype digital employee attendance and working-hours management system for Umzamo Analytical Services that provides a more reliable and verifiable method of recording employee clock-in, clock-out and overtime information

OBJECTIVES:

- Investigate Umzamo Analytical Services existing attendance process
- Identify problems associated with manual attendance recording
- Identify the needs of employees, supervisors and management
- Define a functional and non-functional system requirements
- Design a realistic employee attendance management solution
- Develop a Flutter mobile application for employee attendance function
- Develop an ASP.NET application/API for backend and administrative functions
- Use Supabase as the shared database/backend
- Automatically record clock-in and clock-out timestamps
- Calculate working hours from recorded attendance data
- Provide appropriate attendance reports
- Test the system against confirmed stakeholder requirements
- Evaluate whether the proposed system addresses the identified problem

1.6 Project Scope

Employee Functions	**Management Functions**	**Administration Functions**	**System Functions**
○ Employee clock-in	○ View employee attendance	○ Manage employee account	○ Role based access
○ Employee clock-out	○ Search attendance records	○ Manage attendance records	○ Automatic timestamp generation
○ Automated timestamp recording	○ Filter attendance records	○ Handle attendance corrections	○ Working hours calculation
○ View current attendance status	○ Monitor attendance	○ Generate attendance reports	○ Attendance record storage
○ View attendance history	○ Review overtime		○ Overtime management
○ View recorded working hours	○ Approve or reject overtime		○ Audit information for changes

The project may be constrained by:

- Available semester time
- Group members technical skills
- Stakeholder availability
- Academic assessment deadlines

Project Assumptions:

- Stakeholders will provide information about the existing attendance process
- The organisation has a defined method of identifying employees
- The group can obtain sufficient technical knowledge to use Flutter, ASP.NET and Supabase
- The prototype will not process actual salary payments
- The organisation will provide feedback on the proposed solution

1.7 Deliverables and acceptance criteria

Deliverable	**Acceptance criteria**
Flutter mobile application	Employees can successfully log in, clock in and clock out
ASP.NET backend	API successfully processes and validates attendance requests
Supabase	Attendance records stored correctly
Attendance management	Authorised users can view attendance history
Reports	Overtime can be recorded and managed according to approved rules

1.8 Requirements Prioritisation

Requirements	Priority
User authentication	Must have
Employee Clock-In	Must have
Automatic Clock- In timestamp	Must have
Employee Clock-Out	Must have
Automatic Clock-out timestamp	Must have
Attendance history	Must have
Working- hour calculations	Must have

Management attendance monitoring	Must have
Overtime management	Must have
Employee management	Must have

1.9 Stakeholder Needs

Employees	Management	Administration	Finance
○ Simple clock-in and clock-out process	○ Reliable attendance information	○ Centralised attendance records	○ Accurate salary based on working hour information
○ Ability to view their attendance	○ Attendance reports	○ Reduced reliance on paper forms	
○ A reliable method		○ Reliable attendance information	

1.10 Functional Requirements:

ID	Functional Requirements

FR1	The system must allow authorised users to securely log into the system.
FR2	The system must allow an authenticated employee to clock in.
FR3	The system must automatically record the date and time when the employee clocks in.
FR4	The system must allow an authenticated employee to clock out.
FR5	The system must automatically record the date and time when the employee clocks out.
FR6	The system must allow employees to view their recorded attendance history.

FR7	The system must calculate working hours using the recorded clock-in and clock-out timestamps.
FR8	Authorised supervisors and management users must be able to view employee attendance records.
FR9	The system must provide a mechanism for recording and managing overtime.
FR10	Authorised administrators must be able to manage employee information.

1.11 Non- Functional Requirements

ID	Non-Functional Requirements
NFR1	The system should be simple and intuitive so that employees can clock in and out with minimal interaction
NFR2	The system should process normal clock-in and clock-out requests within an acceptable response time under normal operating conditions
NFR3	Only authorised users should be able to access employee attendance information
NFR4	Employee information and attendance records must be protected from unauthorised access
NFR5	The system should be reliable to store clock-in and clock-out information
NFR6	The system should be available all the time especially during working hours

1.12 System Requirements

User Requirements	Software	Hardware
○ Employee account	○ Flutter mobile application	○ Android smartphone /device for employees
○ Management /admin account	○ ASP.NET backend/API	○ Computer for management and administration
○ Appropriate authentication credentials	○ Supabase database/backend	○ Internet/network connection
	○ Supported web browser for management functions	

1.13 Integration Requirements

The Sebenza Clocking System will integrate these three main technologies

1. **Flutter:** Flutter will be used to develop the employee mobile application/ system. The application will communicate with the backend to,
   - Authenticate users
   - Submit clocking- in events
   - Submit clocking- out events
   - Retrieve attendance information
   - Display working hours and overtime hours

2. **ASP.NET:** ASP.NET will provide the server-side applications/ API. The asp.net component will,
   - Process requests from the Flutter application
   - Apply business rules
   - Validate attendance information
   - Manage authentications
   - Communicate with supabase
   - Provide necessary attendance and overtime information

3. **Supabase:** Supabase will provide the shared backend which will contain relevant information such as,
   - Users
   - Employees
   - Departments
   - Attendance
   - Overtime
   - Attendance corrections

1.14 Requirements change log

Version	Requirement Changes	Reason	Date
01	Initial Sebenza Clocking System concept	Identified manual attendance process	**18 August 2026**
02	Automatic clock-in timestamps added	Stakeholders identified inaccurate manual clock-in times	**22 August 2026**
03	Automatic clock-out timestamps added	Improved reliability of recorded clock-out times	**22 August 2026**
04	Overtime approval process proposed	Stakeholders identified inaccurate overtime records	**29 August 2026**

1.15 Refined Project Scope

Following the initial stakeholder investigation, the refined project scope is centered on creating a digital attendance system that records employee clock-in and clock-out events using automatically generated timestamps

The project will not initially include payroll processing, biometric hardware, facial recognition or external payroll integration.

The core system will include:

- Clock-in
- Clock-out
- Automatic timestamps
- Attendance history
- Working hours calculations
- Attendance monitoring
- Overtime management
- Attendance
- Basic reporting

1.16 Feasibility Study

Technical Feasibility Area: The proposed system/ application will use Flutter, ASP.NET and Supabase. Potential technical challenges include,

- API development
- Database design
- Flutter and ASP.NET communication
- Supabase configuration
- Role-based access control

Operational Feasibility Area: This proposed system/ application will improve the reliability of attendance records by automatically recording clocking- in and clocking- out time. However, this feasibility will depend on,

- The willingness of the employee to use the system
- Management acceptance
- Workplace connectivity

Economic Feasibility Area: The proposed system doesn't require any physical clocking hardware. Meanwhile potential costs may be,

- Internet connectivity
- Devices for developing the system
- Future production deployment

Schedule Feasibility Area: The system must be developed within the semester. Therefore, the following core functions will be prioritized,

- Clock- in
- Clock-out
- Attendance records
- Working hours calculations
- Overtime management
- Reporting

Legal Feasibility Area: The system may process employee personal information and attendance information. Therefore, the system will consider the following,

- Privacy
- Access control
- Data protection
- Secure storage

Ethical Feasibility Area: Attendance information will affect employees so the system should,

- Record information accurately
- Preventing unauthorized modification
- Provide an appropriate
- Protect private employee information

Overall Feasibility Decision: The Sebenza Clocking System is considered feasible as an academic prototype. The project will prioritise it’s core functionalities before considering additional features

1.17 Project Risk Register

Risk ID	Risk Statement	Rating	Impact	Risk Level	Mitigation	Contingency	Risk Owner	Risk Status
Risk ID: 01	Stakeholder becomes unavailable	3-Possible	High	High	Schedule meetings early	Use remote communication	Stakeholder	Open
Risk ID: 02	Requirement change after development t begins	4-Likely	High	High	Confirm requirements early	Update requirements	Group members	Open
Risk ID: 03	Flutter and ASP.NET integration problems	4-Likely	High	High	Test API early	Simplify integration	Group members	Open
Risk ID: 04	Supabase problems	4-Likely	Medium	Medium	Develop supabase early	Rebuild affected components	Group members	Open
Risk ID: 05	System records incorrect attendance	3-Possible	High	High	Validate timestamps and logic	Provide authorised correction process	Group members	Mitigated
Risk ID: 06	Development exceeds available time	4-Likely	High	High	Monitor Microsoft Project plan and progress tracker	Prioritise core MVP	Group members	Open

Risk ID 07	Employee or management have a difficult time adapting to new digital system	possible	high	high	Keep the interface simple, provide basic user guidance, and demonstrate the system to stakeholders before evaluation	Provide additional guidance, simplify the interface where possible and collect user feedback.	Group members	open
Risk ID 08	Attendance records may be lost or become unavailable because of a database or backend failure.	possible	High	High	Use Supabase's database features appropriately, test data storage/retrieval, and maintain backups or recovery procedures where possible.	If attendance data becomes unavailable, investigate the database issue, restore available data from a backup/recovery point, and temporarily record attendance manually until the system is restored.	Group members	open

Three Identified as Highest Risk

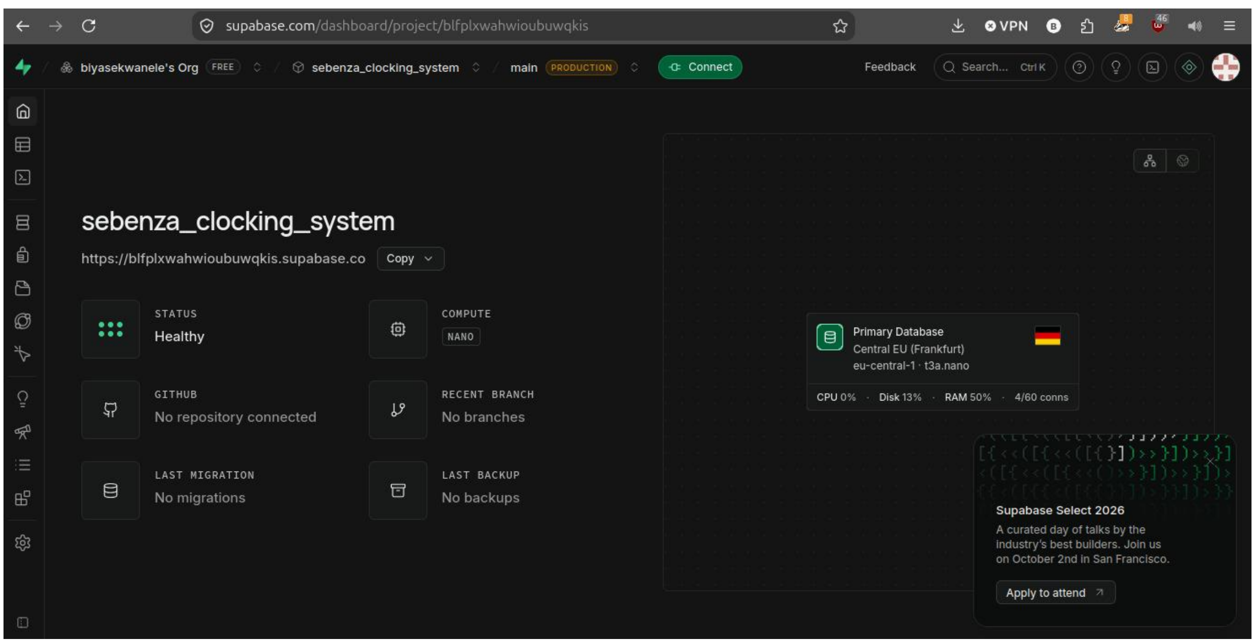
Excessive Project Scope: The project might become too large if the team attempts to implement advanced features like biometrics, facial recognition and complex analytics

Integration Problems: The system will require Flutter, ASP.NET and Supabase to communicate correctly

Unauthorised Access: Attendance information is potentially sensitive employee information

GitHub Repo Link: https://github.com/biyasekwanele/sebenza-clocking-system

Supabase Screenshot



Supporting evidence

SEBENZA CLOCKING SYSTEM

Purpose of the interview

The purpose of this interview is to gather information from Umzamo Analytical Services about the current employee attendance and working hours process, the problems experienced with the existing manual paper-based system, and the requirements for the proposed Sebenza Clocking System.

STAKEHOLDER INTERVIEW QUESTIONS

1. How do employees currently record their clock-in and clock-out times?
2. How are attendance records stored and maintained?
3. What are the main problems you experience with the current paper-based attendance system?
4. Have there been situations where employees recorded an incorrect clock-in or clock-out time?
5. Should employees be able to view their attendance history?
6. Should the system automatically record the exact date and time when an employee clocks in and out?
7. How should the system calculate the total working hours from the clock-in and clock-out times?
8. What security measures should be used to protect employee attendance information?
9. Who should be authorised to access employee attendance records?
10. Would biometric or facial-recognition features be useful for the organisation in the future?

Figure 1 Interview Questions

Requirements Questionnaire Response

Organisation: Umzamowonke Trading and Project t/a Umzamo Analytical Services (UAS)

Respondent: Wendy Pako

Position: Human Resources (HR) Representative

Date: 31 August 2026

The following responses were provided by Wendy Pako, the HR representative of Umzamowonke Trading and Projects t/a Umzamo Analytical Services. The responses describe the organisation's current employee attendance process and the requirements for the proposed attendance management system.

Answers

1. **How do employees currently record their clock-in and clock-out times?**
   Employees manually record their clock-in and clock-out times in a timesheet register. After recording their times, they sign the register to confirm and authorise the information.
2. **How are attendance records stored and maintained?**
   Attendance records are stored in physical timesheet registers. The registers are kept and maintained by the company, with employees signing their entries to confirm that the information is correct.
3. **What are the main problems you experience with the current paper-based attendance system?**
   The current system is time-consuming and requires manual recording and calculations. Timesheet registers can also be lost, damaged, or difficult to search. There may also be problems with unclear handwriting, missing entries, and incorrect calculations.
4. **Have there been situations where employees recorded an incorrect clock-in or clock-out time?**
   Yes. Employees may sometimes record an incorrect time or forget to record their

Figure 2 Interview Answers

   occur and may require corrections and authorisation.
5. **Should employees be able to view their attendance history?**

Yes. Employees should be able to view their attendance history to check their recorded working hours and ensure that their information is accurate.

6. **Should the system automatically record the exact date and time when an employee clocks in and out?**

   Yes. The system should automatically record the exact date and time when an employee clocks in and clocks out. This would reduce manual errors and provide more accurate attendance records.

7. **How should the system calculate the total working hours from the clock-in and clock-out times?**

   The system should automatically calculate the total working hours by subtracting the clock-in time from the clock-out time. Any approved break time should also be deducted where applicable.

8. **What security measures should be used to protect employee attendance information?**

   The system should use usernames and passwords and restrict access according to the user's role. Attendance information should be securely stored, and only authorised users should be allowed to view or modify records.

9. **Who should be authorised to access employee attendance records?**

   Employees should be able to view their own attendance records. Managers, supervisors, or authorised HR/administrative staff should have permission to access and manage employee attendance records. Changes to records should require proper authorisation.

10. **Would biometric or facial-recognition features be useful for the organisation in the future?**

    Yes. Biometric or facial-recognition features could be useful in the future because they could automatically identify employees and reduce the possibility of one employee signing in on behalf of another. However, the organisation would need to consider the cost, privacy, security, and employee consent before introducing these features.

Figure 4 Continuation of Interview answers

Figure 5 Stakeholder engagement

Meeting notes 21/08/2026

Introductions

- 11: 57 speaking to HR lady
- 12:03 pedro explaining what we are about
- 12:09 he askes questions (questionnaire)
- 12:26 Uas still using paper sheet to record. -
- attendance (that's a problem)
- 12:30. Lady suggests she should also send -

answers using the company letter head.

- still in talks of what theyd like to do in future (uas)
- we would like to create a system for them
- agrees to our idea to create and tells us what she would like us to focus on is thats they can verify the times each member cloked in at

Formalities

Goodbye - 12:59

Figure 6 meeting minutes